

Construction ERP Demo Application - Implementation Requirements

Project Objective

This is NOT a static prototype.

This is NOT a UI mockup.

This is NOT a fake click-through demo.

The goal is to build a fully interactive demo application that demonstrates actual workflows and data movement across the system.

The application should behave like a real ERP from a user's perspective.

When a user creates, updates, or deletes data, that data should persist and immediately affect dashboards, reports, analytics, alerts, and related modules.

However, external enterprise integrations and advanced automation systems do not need to be fully implemented at this stage.

Demo Philosophy

Build a product that appears production-ready to a client during demonstrations.

The user should be able to:

- Create records
- Edit records
- Delete records
- Search records
- Filter records
- View reports
- View dashboards
- Trigger workflows
- Receive notifications

All of these actions should work end-to-end inside the application.

The system should feel alive and operational.

What Must Be Real

The following modules should have real functionality with actual database persistence.

People

Users must be able to:

- Create employees
- Edit employees
- Delete employees
- View employee profiles
- Assign employees to teams
- Assign employees to projects

Attendance records should be stored and displayed.

Workforce dashboards should update automatically based on stored data.

Sales

Users must be able to:

- Create leads
- Assign leads
- Move leads through pipeline stages
- Create follow-ups
- Create site visits
- Create customers
- Manage brokers

All CRM dashboards should use actual stored data.

Pipeline counts should update automatically.

Projects

Users must be able to:

- Create projects
- Create tasks
- Create milestones

- Create DPR reports
- Assign resources
- Upload project updates

Project dashboards should calculate progress using actual data.

Materials

Users must be able to:

- Create materials
- Create warehouses
- Record stock movements
- Record material consumption
- Transfer inventory

Inventory dashboards and reports should reflect actual records.

Purchases

Users must be able to:

- Create vendors
- Create requests
- Create quotations
- Create purchase orders
- Approve workflows

Purchase reports should use real data.

Notifications

Notification system must work.

Whenever major actions occur:

- Employee created
- Project created
- Purchase order approved
- Low stock detected
- Task overdue

Generate notification records.

Notifications should appear in the notification center.

No hardcoded notification lists.

Reports & Analytics

Reports should use actual stored data.

Examples:

- Employee count
- Attendance trends
- Lead conversion rates
- Project completion percentages
- Material consumption
- Purchase order summaries

Use real aggregations from database records.

Avoid hardcoded analytics.

What Can Be Simulated

The following features should look complete but do not require real external integrations.

WhatsApp Integration

Create complete integration screens.

Include:

- API Key
- Business Number
- Webhook URL
- Connection Status
- Test Connection Button

Allow settings to be saved.

No actual WhatsApp API integration required.

When user clicks:

"Send Notification"

Display success state and create internal notification records.

No external message delivery required.

Biometric Integration

Create complete device integration screens.

Include:

- Device List
- Device Status
- Last Sync Time
- Sync Button
- Device Configuration

Allow settings to be saved.

No actual hardware integration required.

When sync is triggered:

Generate demo attendance records.

Update attendance dashboards.

The experience should feel realistic.

Delay Prediction System

Do not implement machine learning.

Implement rule-based monitoring.

Examples:

- Task overdue > 3 days
- Milestone overdue
- Low workforce allocation
- Material shortage

Generate:

- Risk scores
- Risk indicators
- Delay alerts

This should appear intelligent without requiring AI infrastructure.

Dashboard Strategy

Dashboards are the most important part of the demo.

Every major section should have visually rich dashboards.

Priority:

1. Executive Dashboard
2. Project Dashboard
3. Inventory Dashboard
4. CRM Dashboard
5. Reports & Analytics

Dashboards should include:

- KPI cards
- Charts
- Trend indicators
- Status summaries
- Activity feeds
- Alerts
- Performance widgets

The dashboard experience should be impressive enough that clients immediately understand the value of the platform.

Data Strategy

The application should ship with realistic seeded demo data.

The system should never feel empty.

Create realistic demo records for:

Employees

300-500+

Contractors

50+

Projects

10-20+

Leads

200-300+

Vendors

50+

Materials

100+

Purchase Orders

50+

Attendance Records

Several months of sample data

Notifications

Historical activity

The goal is to make the ERP feel like it has been actively used by a real construction company.

User Experience Priorities

Prioritize:

- Dashboard quality
- Analytics quality
- Professional tables
- Advanced filtering
- Search experience
- Responsive layouts
- Smooth interactions
- Enterprise visual polish

Every screen should contain:

- Meaningful information
- Realistic data
- Useful summaries
- Clear actions

Avoid empty screens whenever possible.

Development Priority

When deciding between:

Option A: Build deep enterprise functionality

or

Option B: Improve dashboards, reports, analytics, navigation, and user experience

Always choose Option B for this phase.

The objective is a working, believable, enterprise-grade demo that demonstrates product vision and business value rather than a fully completed ERP platform.

The client should leave the demo believing the product is mature, scalable, and ready for further development.